How to Write an Education Research Report

A Research Report

A practical guide for teachers, student-teachers and school leaders: asking a clear question, gathering honest evidence in a real classroom, analysing it simply, and writing it up so that other teachers can learn from it.

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This report is an original guide written for classroom practitioners. All examples are illustrative and fictional unless stated otherwise. No part of this report reproduces another publication.

Abstract

Education research is the disciplined study of teaching and learning. It asks a clear question about what happens in classrooms, gathers evidence in a careful and honest way, and reports what was found so that practice can improve. This report explains how a teacher — not only a university researcher — can carry out that work and write it up as a complete research report.

The report begins by defining education research and explaining why teachers need it: to replace guesswork and habit with evidence, to test whether a new method really works, and to share useful findings with colleagues. It then describes the main types of education research — experimental, survey, case study, ethnographic and action research — and explains when each type is appropriate, what it can show, and what it cannot show.

The central method discussed is action research, the Plan–Act–Observe–Reflect cycle that fits naturally into a teacher's daily work. The report shows how to turn a classroom worry into a focused research question, how to collect data with tests, questionnaires, interviews and observation, and how to analyse that data simply and honestly, using counts, percentages and clear comparisons rather than complicated statistics. It gives the standard structure of a research report — abstract, introduction, literature review, methodology, findings, discussion, recommendations — and practical guidance on writing findings as evidence, not opinion.

A complete worked example is included: a clearly labelled fictional action-research study on improving reading fluency in a Grade 5 class, written out as a short model report. The report closes with guidance on using findings to improve teaching, ethical rules for research with children, common mistakes to avoid, and a final checklist that a writer can use before submitting a report. The aim throughout is the same: research that is small, honest and useful is far better than research that is large, vague and unused.

Keywords: education research; action research; classroom research; research report writing; teacher inquiry; reading fluency; research ethics.

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How to use this report

Read Chapters 1 to 4 first if you have never done research before. If you already have a question in mind, start at Chapter 4 and use Chapters 5 and 6 while you collect and analyse your data. Chapter 11 is a model you can imitate in structure — not in content. Your own study will have different pupils, different numbers and different findings, and that is exactly as it should be.

Chapter 1

Introduction — What Education Research Is and Why Teachers Need It

Every teacher makes decisions all day: which example to use, how long to spend on a topic, whether to repeat a lesson, how to help a pupil who is struggling. Most of those decisions are based on experience, habit, or what a colleague suggested. Education research is a more careful way of making the same decisions. It means asking a clear question, collecting evidence that can answer it, thinking honestly about what the evidence shows, and writing the result down so that others can check it and learn from it.

A simple definition is useful: education research is systematic inquiry into teaching, learning, schools and the people in them. Three words in that definition matter. Systematic means the work follows a plan rather than relying on memory or a single impression. Inquiry means it starts with a genuine question, not with an answer already decided. Evidence means the conclusion rests on something that was observed, counted or recorded, not only on what the researcher hoped to find.

Why teachers need research

First, research protects teachers from being misled by their own impressions. A lively lesson can feel successful even when pupils learned very little, and a quiet lesson can hide real learning. A short test before and after a new method, or a careful look at pupils' written work, gives a much better guide than a feeling. Second, research helps a school spend its limited time and money wisely. If a new reading programme, a new textbook or a new use of class time does not improve learning, it is better to know that in one term than to continue it for three years. Third, research gives teachers a professional voice. When a teacher can say, 'I tried this with my class of thirty-two pupils for six weeks, and here are the results,' that claim carries far more weight with a head teacher, a parent or an education officer than a general opinion.

Research also matters because teaching is full of claims that sound convincing but are not true for every class. A method that worked brilliantly in one school may fail in another because the pupils, the class size, the language of instruction or the available time are different. Small, local research tests whether a general idea works in a particular place, with particular children, under particular conditions. That local test is often more useful to a school than a large study carried out far away.

Research is not only for universities

Many teachers believe research belongs to universities and that they themselves are only supposed to apply what others have discovered. That belief wastes the best source of knowledge about classrooms: the teachers who work in them every day. A

teacher-researcher sees things an outside researcher cannot see — the pupil who reads well at home but freezes in class, the lesson that fails on Mondays but works on Wednesdays, the group that needs five more minutes. When teachers record those observations carefully, they produce knowledge that no outsider could produce.

Teacher research does not need expensive equipment, large samples or advanced statistics. It needs a clear question, a simple plan, honest recording, and the willingness to report what actually happened — including results that were disappointing. A small study done well is worth more than a large study done carelessly. The rest of this report explains how to do the small study well, step by step, and how to write it up so that it can be read, understood and trusted.

A note on honesty

The most important quality in research is not cleverness but honesty. Report what you did, exactly as you did it. Report what you found, even when it contradicts what you expected. A study that finds 'no improvement' is not a failure — it is useful information that saves other teachers from repeating the same approach.

Chapter 2

Types of Education Research

There is no single way to do education research. Different questions need different methods, and a good researcher chooses the method that fits the question rather than forcing the question to fit a favourite method. This chapter describes five common types, with their strengths, their limits, and examples of the questions each one answers best.

2.1 Experimental research

Experimental research tests whether one thing causes another. The researcher changes one factor — for example, a teaching method — while keeping other factors as similar as possible, and then measures the result. In its strongest form, pupils are assigned at random to two groups: one group receives the new method and the other continues with the usual method, and both groups take the same test afterwards. If the first group scores clearly higher, and the groups were similar to begin with, the researcher has reasonable evidence that the method caused the improvement.

True experiments are powerful but difficult in schools. It is often unfair or impractical to give one class a better method and deny it to another, and classes are rarely identical. Teachers therefore often use a weaker but still useful version: they measure their own class before and after a change, and compare the results. This shows change over time, but it cannot prove that the change caused the improvement, because other things — extra practice, growing maturity, a different test — may also have contributed. That limit must be stated honestly in the report.

2.2 Survey research

Survey research asks many people the same questions, usually through a questionnaire, and describes what they think, know or do. A survey can show, for example, how many parents help their children with homework, how teachers feel about a new curriculum, or how often pupils read at home. Its strength is breadth: it can cover hundreds of people quickly and cheaply. Its weakness is depth: a questionnaire can record what people say, but it cannot always show what they actually do, and people sometimes give the answer they think is expected rather than the true one. Good survey questions are short, neutral and unambiguous, and they are tested on a few people before the survey begins.

2.3 Case study research

A case study looks in depth at one person, one class, one school or one event. Instead of covering many people briefly, it studies a few in great detail, using several kinds of evidence — test scores, interviews, observation notes, samples of work — and builds a rich picture of what happened and why. A case study of one struggling

reader, followed carefully for a term, can reveal more about reading difficulty than a survey of a hundred pupils. The limit is that one case may not represent others. A case study can suggest an explanation and generate ideas, but it cannot prove that the same explanation applies everywhere.

2.4 Ethnographic research

Ethnography comes from anthropology. The researcher spends a long time inside a school or classroom community, observing daily life, listening to how people talk, and trying to understand the culture — the unwritten rules, values and routines that shape behaviour. An ethnographic study might explore how new pupils learn the rules of a school, how teachers talk about ability in the staffroom, or how children use language in the playground. It produces deep understanding, but it takes months, it depends heavily on the researcher's interpretation, and its findings are tied to one particular setting. For most teacher-researchers, full ethnography is impractical, but its central habit — careful, patient observation without rushing to judge — is valuable in every kind of study.

2.5 Action research

Action research is research carried out by practitioners to improve their own practice. A teacher identifies a problem in her own classroom, tries a change, observes what happens, reflects on the results, and then tries again with a better plan. It is practical, small-scale and cyclical, and it is the main method of this report, described in detail in Chapter 3. Its strength is direct usefulness: the findings feed immediately back into teaching. Its limit is that the researcher is also the teacher, so objectivity must be guarded carefully — by keeping records, by using more than one kind of evidence, and by reporting honestly.

Choosing a type

Ask first: what do I want to know? To test whether a method works, use an experiment or action research. To describe what many people think or do, use a survey. To understand one situation deeply, use a case study. To understand a culture over time, use ethnography. The question chooses the method — never the other way round.

Chapter 3

Action Research for Teachers: The Plan–Act–Observe–Reflect Cycle

Action research was developed for people who want to improve their own work while they are doing it. Its best-known form is a repeating cycle with four stages: plan, act, observe and reflect. A teacher goes round the cycle once, learns from the result, and then begins again with a better plan. Two or three cycles in a term are usually enough to produce a useful study and a genuine improvement in teaching.

Stage 1 — Plan

Planning starts with a problem the teacher can see in her own classroom: pupils who cannot read a passage fluently, homework that is rarely completed, girls who seldom speak in class discussions, or a topic that most pupils fail every year. The teacher narrows the problem to something specific and observable, reads a little about how others have tackled it, and designs a practical change that can be tried within the available time and resources. The plan also states how the change will be judged: what evidence will be collected, and what would count as improvement. A plan written down before the work starts is itself a research document; it prevents the teacher from changing the goal halfway through to suit the results.

Stage 2 — Act

Acting means carrying out the planned change in the real classroom, for a fixed period — typically four to eight weeks. The change should be clear enough that another teacher could repeat it: for example, ten minutes of paired reading at the start of every English lesson, with a simple record sheet for each pair. During this stage the teacher teaches normally, except for the planned change, and keeps a brief diary of what happened each day: what worked, what went wrong, and any surprises. A diary entry of five lines a day is far more useful than a long account written from memory at the end of the term.

Stage 3 — Observe

Observation is the systematic collection of evidence while the action is running. It is not the same as casual noticing. The teacher collects the evidence named in the plan: a reading test before and after, a count of how many pupils completed homework, observation notes on who spoke in discussions, or short interviews with a few pupils. Evidence should come from more than one source wherever possible, so that a weakness in one source is balanced by another. If test scores improve but pupils say they hated the activity, both facts must be reported.

Stage 4 — Reflect

Reflection is the stage where the teacher stops, looks at all the evidence together, and asks: what happened, why did it happen, and what should I do next? Reflection is honest when it considers alternative explanations. Perhaps scores improved because pupils had more practice, not because of the new method. Perhaps only the stronger pupils improved. Perhaps the method worked but took too much time to be practical. The reflection leads to a decision: continue the change, modify it, or abandon it — and that decision begins the next cycle, with a revised plan.

Worked example of one cycle

Problem: Grade 5 pupils read aloud slowly and make many errors. Plan: ten minutes of paired reading, four days a week, for six weeks; measure fluency with a one-minute reading test before and after. Act: the teacher runs the paired reading and keeps a diary. Observe: pre-test and post-test scores, diary notes, and three short pupil interviews. Reflect: average fluency rose from 62 to 78 words per minute, but five pupils showed no gain; the teacher decides to add extra support for those five in the next cycle. Chapter 11 writes this example out as a full model report.

Why the cycle matters

The cycle matters because teaching is never finished. A single experiment gives a single answer; a cycle gives a habit. Teachers who practise action research become more observant, more willing to test their own assumptions, and more able to explain to others why they teach as they do. Over a career, that habit is worth more than any single study. It also builds a school culture in which improvement is normal: teachers share findings in staff meetings, try each other's ideas, and build a shared stock of local knowledge about what works with their pupils, in their school, under their conditions.

Action research has limits, and a good report states them. The teacher is both the researcher and the person being judged, so bias is a real risk. The sample is one class, so findings may not apply elsewhere. The time is short, so long-term effects cannot be seen. Stating these limits does not weaken a report — it strengthens it, because it shows the reader exactly how far the findings can be trusted.

Chapter 4

Formulating a Research Question in a Classroom Setting

A research study succeeds or fails on its question. A vague question produces vague data and a vague report. A clear, focused question guides every later decision: what data to collect, whom to study, how to analyse the results, and what the report should conclude. Formulating the question is therefore the most important step in the whole study, and it deserves more time than most beginners give it.

From a worry to a question

Most studies begin as a worry: 'My pupils cannot write a paragraph,' 'Homework is never done,' 'Boys dominate class discussion.' A worry is a good starting point, but it is not yet a research question. It must be narrowed in three ways. First, it must name the pupils: which class, which age, how many. Second, it must name the change or the topic precisely: not 'better writing' but 'writing a five-sentence paragraph with a topic sentence'. Third, it must name what will count as evidence: a test, a count, a sample of work. A question that has been narrowed in these three ways can actually be answered.

What makes a good research question

A good classroom research question is clear, focused, researchable and useful. Clear means that another teacher reading it would understand exactly what is being asked. Focused means it asks about one thing, not five things at once. Researchable means it can be answered with evidence that the teacher can realistically collect in the time available — a question that needs a year and a hundred schools is not researchable for one teacher in one term. Useful means that the answer will change what the teacher does: if the answer would make no difference to practice, the question is not worth asking.

Good questions are usually open rather than closed. 'Do pupils like group work?' can be answered with a single word and tells the teacher very little. 'How does paired reading affect the reading fluency of Grade 5 pupils over six weeks?' invites evidence, measurement and discussion. Good questions also avoid words that cannot be measured, such as 'better', 'good' or 'effective', unless the writer defines exactly what those words mean in this study.

Examples: weak questions made strong

1. Weak: 'How can I improve English?' Strong: 'Does ten minutes of daily paired reading improve the reading fluency of Grade 5 pupils over six weeks, measured by words read correctly per minute?'

2. Weak: 'Why do pupils not do homework?' Strong: 'What reasons do Grade 6 pupils give for not completing mathematics homework, and how do those reasons differ between boys and girls?'

3. Weak: 'Is group work good?' Strong: 'How does structured group work, compared with individual seatwork, affect the number of pupils who complete a science task correctly in Grade 7?'

4. Weak: 'How can I make my class disciplined?' Strong: 'Does a clear seating plan and a two-minute settling routine reduce the time taken to start lessons in Grade 4, measured over four weeks?'

Each strong version names the pupils, the change or topic, the time period, and the evidence. Notice also that the strong questions are honest about what they can show. The first question asks whether paired reading improves fluency in one class over six weeks; it does not claim to prove that paired reading works for all children everywhere. Modesty in the question is a strength, not a weakness, because it makes the study achievable and its findings trustworthy.

Sub-questions and a working title

A main question can be supported by two or three sub-questions that break it into parts. For the paired-reading study, sub-questions might be: How much does fluency change for the weakest readers? How do pupils feel about paired reading? How much class time does the activity require? Sub-questions keep the study organised and give the findings section a clear structure — one section per sub-question. A working title can be drafted from the question, for example: 'Improving Reading Fluency Through Paired Reading: An Action Research Study in Grade 5'. The title can be refined at the end, when the findings are known.

Test your question

Before collecting any data, ask four questions: Can I explain this question to a colleague in one sentence? Can I collect evidence for it in the time I have? Will the answer change my teaching? Could the answer surprise me? If the answer to any of these is no, revise the question before you begin.

Chapter 5

Data Collection in Schools: Tests, Questionnaires, Interviews and Observation

Data are the evidence on which a research report stands. Good data are collected deliberately, recorded at the time, and kept safely. Bad data — remembered weeks later, collected with a confusing instrument, or recorded selectively — can ruin an otherwise good study. This chapter describes the four methods a classroom researcher will use most often, and compares them so that a teacher can choose wisely and combine methods to strengthen a study.

5.1 Tests

Tests give numbers that can be compared before and after a change. A reading fluency test, a spelling test or a short mathematics quiz can show whether learning has improved, and by how much. To be useful, a test must be fair: the pre-test and the post-test should be of similar difficulty, pupils should have the same amount of time, and the marking should follow a fixed scheme so that the same answer always receives the same mark. A test that is too easy or too hard tells the researcher nothing, because every pupil scores at the top or at the bottom. It is good practice to try the test on a few pupils first and to keep a copy of the test and the marking scheme with the report, so that a reader can judge its quality.

5.2 Questionnaires

A questionnaire asks many people the same written questions and is useful for finding out what pupils, parents or teachers think, feel or do. A good questionnaire is short — rarely more than ten or twelve questions for pupils — uses simple language, and avoids leading questions that push the respondent towards one answer. 'Do you agree that paired reading is the best way to learn?' is a leading question; 'How did you feel about paired reading?' with a choice of answers is better. Closed questions with fixed choices (for example, 'Every day / Sometimes / Never') are easy to count. Open questions ('What did you like most about paired reading?') give richer answers but take longer to analyse. Questionnaires for young children should be read aloud and kept very short, and the researcher should explain that there are no right or wrong answers.

5.3 Interviews

An interview is a guided conversation. It gives deeper information than a questionnaire, because the researcher can ask follow-up questions: 'You said reading is hard — can you tell me which part is hardest?' Interviews work best with a small number of people — perhaps five or six pupils — chosen to represent different levels of ability or different opinions. A short list of open questions, prepared in advance,

keeps the interview focused without making it rigid. The researcher should take notes during the interview or, with permission, record it, and should write up the notes on the same day, while the conversation is still fresh. Quotations should be recorded in the pupil's own words wherever possible, and the report should say how many pupils were interviewed and how they were chosen.

5.4 Observation

Observation means watching and recording what actually happens, rather than relying on what people say happens. A teacher might observe how many pupils raise their hands, how long it takes a class to settle, or how pupils behave during group work, using a simple tally sheet prepared in advance. Observation is powerful because it captures real behaviour, but it is also the method most open to bias: a researcher who expects an improvement may unconsciously notice the examples that confirm it and miss the ones that do not. Two safeguards help: decide in advance exactly what will be counted, and, where possible, ask a colleague to observe the same lesson and compare notes. Observation notes should record facts ('Seven pupils spoke; three spoke more than once') rather than judgements ('The class was lively').

5.5 Comparing the methods

No single method is best for every purpose. The table below compares the four methods on the points that matter most to a teacher-researcher: what each method shows, how many people it can cover, how long it takes, and its main weakness. In practice, good studies combine two or three methods, so that the weakness of one is covered by the strength of another. This is called triangulation: if test scores, interviews and observation all point in the same direction, the finding is much more convincing than if it rests on one method alone.

Method	Best for	Coverage	Main weakness
Tests	Measuring learning before and after a change	Whole class	Miss feelings and reasons
Questionnaires	Views of many pupils or parents quickly	Many people	Answers may be what is expected
Interviews	Reasons, feelings and detail in pupils' own words	Few people	Slow; small sample
Observation	What pupils actually do in class	One class at a time	Observer bias is a risk

Whatever methods are chosen, three practical rules apply. First, pilot the instrument: try the questionnaire or test on a few pupils and fix the confusing questions before the real study begins. Second, keep everything: the blank questionnaire, the test papers, the tally sheets and the interview notes are the raw evidence of the study and should be stored safely, with names removed where confidentiality requires it.

Third, collect the same kind of data in the same way at the start and at the end, so that the two sets of results can fairly be compared.

Chapter 6

Analysing Classroom Data Simply and Honestly

Many teachers are put off research by the fear of statistics. In fact, most classroom studies need only simple analysis: counts, percentages, averages and clear comparisons. The aim of analysis is not to impress the reader with complicated mathematics, but to show plainly what the evidence says — and what it does not say. This chapter explains how to do that, for both numbers and words.

6.1 Working with numbers

Begin by organising the numbers in a table. For a pre-test and post-test study, list each pupil's score before and after, and the difference. Then calculate a few simple summaries. The average (mean) shows the typical score. The range — the lowest and highest scores — shows how spread out the results are. Percentages show proportions clearly: '18 of 30 pupils (60%) improved by ten or more words per minute' is clearer than a list of thirty numbers. A simple bar chart, drawn by hand or in a spreadsheet, often shows a pattern more clearly than a paragraph of text. Every table and chart in a report needs a title and a short explanation of what the reader should notice in it.

Two cautions are essential. First, a small change in a small class may be due to chance. If the average rises by two points in a class of thirty, that may mean very little; if it rises by fifteen points, it probably means something. A report should give the actual numbers, so that the reader can judge for herself, rather than claiming a 'big improvement' without showing the figures. Second, averages can hide important differences. If the class average rises but the weakest pupils do not improve at all, that fact must be reported. It is often more useful to show results in groups — for example, the weakest third, the middle third and the strongest third — than to report a single average for the whole class.

6.2 Working with words

Interview answers, open questionnaire responses and observation notes are words, not numbers, and they need a different kind of analysis. The usual approach is to read through all the responses, look for repeated ideas, and group them into themes. For example, when pupils are asked what they liked about paired reading, their answers might fall into themes such as 'help from a partner', 'less fear of mistakes', and 'reading more books'. The report can then say how many pupils mentioned each theme and give one or two short quotations to illustrate it. Quotations make a report vivid, but they must be chosen fairly: a quotation that supports the researcher's view should not be presented as if every pupil said the same thing.

Counting themes is a useful bridge between words and numbers. A table that shows 'Theme — Number of pupils who mentioned it' gives the reader a clear picture and

prevents the researcher from being misled by one memorable comment. The researcher should also look for answers that do not fit the pattern — the pupil who disliked the activity, the unexpected problem — and report them. A report that mentions only the positive findings is not honest, and an experienced reader will notice the omission.

6.3 Staying honest

Honest analysis means reporting what the data show, not what the researcher hoped they would show. It means stating the limits: the class was small, the time was short, the test may have been easier the second time, the researcher was also the teacher. It means not exaggerating small differences, not hiding results that contradict the main finding, and not claiming that a change caused an improvement when the evidence only shows that the two happened at the same time. A modest claim that is supported by evidence is far more valuable than a bold claim that is not. The discussion chapter of the report is the place to consider alternative explanations openly, and to say what further study would be needed to test them.

A simple analysis checklist

Have I shown the actual numbers, not only my opinion of them? Have I shown the range, not only the average? Have I reported results for weaker pupils as well as stronger ones? Have I counted themes in the words, and reported views that disagree with mine? Have I stated the limits of my data? If the answer to each is yes, the analysis is honest.

Chapter 7

Structure of an Education Research Report

A research report follows a standard structure that readers expect. The structure is not a formality: it helps the writer organise the work and helps the reader find what matters quickly. A teacher who is used to the structure can read another teacher's report in a few minutes and judge its quality. This chapter describes each part in order, with guidance on what belongs in it and how long it should be.

Title page and abstract

The title page carries the title, the writer's name, the institution or school, and the date. The title should name the topic, the method and the setting, for example: 'Improving Reading Fluency Through Paired Reading: An Action Research Study in Grade 5'. The abstract, usually 150 to 250 words, is a complete summary in miniature: the question, the method, the main findings, and the main recommendation. Many readers will read only the abstract, so it must be able to stand alone. It is usually written last, after the rest of the report is finished.

Introduction

The introduction explains what the problem is, why it matters, and what the study set out to do. It gives the background of the school and class, describes the problem in concrete terms, states the research question, and explains why the question is worth answering. It should be long enough to make the study understandable to a reader who does not know the school, but short enough that the reader reaches the method without losing interest — usually one to two pages in a short report.

Literature review

The literature review summarises what is already known about the topic. It does not need to be long in a teacher's report: a few paragraphs, drawing on books, articles or official guidance the teacher has actually read, are enough. Its purpose is to show that the study builds on existing knowledge rather than starting from nothing, and to explain why the chosen approach was expected to work. Every source mentioned must be listed at the end of the report, and every source listed must have been mentioned in the text. Inventing references, or listing sources that were never read, is a serious breach of research honesty.

Methodology

The methodology chapter is the heart of the report's trustworthiness. It must describe the study so clearly that another teacher could repeat it. It states who took part and how they were chosen, what was done and for how long, what data were collected and how, and how the data were analysed. It also states the ethical steps taken:

permission, consent and confidentiality. If the methodology is vague, the reader cannot judge whether the findings are reliable, however interesting they may be.

Findings, discussion and recommendations

The findings chapter presents the results, organised by research question or sub-question, with tables, charts and quotations as evidence. It reports what was found and does not yet argue about what it means. The discussion chapter interprets the findings: what they mean, how they compare with what others have found, why unexpected results may have occurred, and what the limits of the study are. The recommendations chapter says what should happen next — for the teacher, for the school, and for further research — and each recommendation should follow from a finding, not from a general wish. A report that recommends something its own findings do not support has failed at the last step.

References and appendices

The reference list gives full details of every source cited. Appendices hold material that supports the report but would interrupt the main text: the questionnaire, the test, the observation sheet, or a sample of pupils' work with names removed. Appendices should be numbered and referred to in the text, so that the reader knows where to find them. A report of the kind described in this guide is typically 3,000 to 6,000 words, plus tables and appendices — long enough to be complete, short enough to be read.

Chapter 8

Writing Findings: Showing Evidence, Not Just Opinions

The findings chapter is where a research report earns its reader's trust — or loses it. Its job is simple: to show what happened, with evidence. Yet it is the chapter where beginners most often go wrong, replacing evidence with opinion, generalisation and enthusiasm. This chapter explains the difference and shows how to write findings that a sceptical reader can believe.

Evidence, not opinion

Consider two sentences. 'Paired reading worked very well and the children loved it.' That is an opinion. It tells the reader what the writer thinks, but gives no reason to agree. Now consider this: 'Average reading fluency rose from 62 to 78 words correct per minute over six weeks. Twenty-four of thirty pupils improved by ten or more words per minute; five showed no gain. In interviews, nine of twelve pupils said they felt less afraid of making mistakes when reading with a partner.' That is evidence. It gives numbers, proportions and the words of pupils, and it reports the pupils who did not improve as well as those who did. A reader may disagree with the writer's interpretation, but she cannot say that no evidence was offered.

Every important claim in a findings chapter should be followed by evidence in the same paragraph: a number, a table, a chart or a quotation. Claims without evidence should be removed or moved to the discussion, where they can be presented as interpretation rather than as fact. Adjectives such as 'very', 'excellent', 'amazing' and 'terrible' are a warning sign: they usually appear where evidence is missing. A report that says 'scores improved by 16 words per minute' needs no adjective; the number speaks for itself.

Organising the findings

Findings should be organised by research question, not by the order in which the data were collected. If the study had three sub-questions, the findings chapter should have three sections, one for each, each with a clear subheading. Within each section, the strongest evidence comes first. Tables and charts should be placed close to the paragraph that discusses them, each with a number and a title, and the paragraph should tell the reader what to notice: 'Table 1 shows that the largest gains were made by pupils who started below 50 words per minute.' A table that is never mentioned in the text should not be in the report.

Writing about numbers and quotations

Numbers should be reported accurately and consistently. If the report says that 24 of

30 pupils improved, it should not say elsewhere that 'most pupils improved' without the number, and it should not round 24 of 30 to 'almost all'. Percentages should be given with the underlying numbers, especially in small classes, because 50% of four pupils is only two pupils. Quotations should be short, in the speaker's own words, and identified only by a code or a role ('Pupil 7', 'a Grade 5 girl'), never by name. A quotation should illustrate a pattern that the numbers already show, not stand alone as the only evidence for a claim.

Example: a findings paragraph done well

'Fluency improved for most pupils. The class average rose from 62 words correct per minute in the pre-test (range 34–91) to 78 in the post-test (range 41–105), an average gain of 16 words per minute (Table 2). Twenty-four pupils gained ten or more words per minute; five gained between zero and four words; one pupil scored two words lower. The largest gains were among pupils who began below 50 words per minute, whose average rose from 43 to 64. In interviews, pupils linked the gain to practice with a partner: as one pupil put it, “When I stop, my partner helps me, so I do not feel afraid.”'

Chapter 9

Using Research Findings to Improve Teaching

Research that is written up and then filed away has failed in its main purpose. The point of classroom research is to change teaching for the better, and the final chapters of a report should make that change as easy as possible for the writer and for other teachers who read it. This chapter explains how findings become action, in the writer's own classroom and in the wider school.

From finding to action in your own classroom

The first use of findings is the most direct: the teacher changes her own practice. If paired reading improved fluency, she continues it, perhaps with extra support for the pupils who did not gain. If it took too much time, she shortens it or moves it to a different part of the lesson. If pupils disliked one part of the activity, she changes that part. The report should record the decision and the reason for it, so that the next cycle of action research begins from a clear position. A useful habit is to write the decision down within a week of finishing the analysis, while the evidence is still fresh, and to fix a date to review it after a further four weeks.

Sharing findings with colleagues

Findings become more valuable when they are shared. A ten-minute presentation in a staff meeting, with one table and one quotation, is often more effective than a long written report that nobody reads. The presentation should answer three questions: What did I try? What happened? What would I recommend? Colleagues should be invited to ask questions and to try the idea in their own classes, and the writer should be honest about what did not work. A school that shares small findings regularly builds a shared memory of what works with its own pupils, which is far more useful than a shelf of unread reports.

When findings should not be generalised

A responsible researcher also knows when not to generalise. A study of one class of thirty pupils over six weeks cannot prove that a method will work in every school, and the report must not claim that it does. Findings can be offered as a suggestion for other teachers to try, with the invitation to test it in their own setting — not as a rule for all schools. Equally, a single study should not be used to justify a large, expensive change across a whole school. A school-wide change needs evidence from more than one class, over more than one term, and preferably from more than one teacher. The report's recommendations should match the strength of its evidence: small evidence, modest recommendations.

Finally, findings should be kept and revisited. A short summary — the question, the method, the main result, the decision — kept in a staff file or a school folder, allows a

new teacher to benefit from work done years earlier. Over time, a school accumulates a practical library of its own research: what helped weak readers, what improved homework completion, what reduced late starts. That library, built slowly from honest small studies, is one of the most valuable things a school can own.

Chapter 10

Ethics in Education Research: Children, Consent and Confidentiality

Research in schools involves children, and children deserve special protection. Ethical rules are not bureaucratic obstacles: they protect pupils from harm, protect parents' right to decide, and protect the researcher from doing something she would later regret. A report that ignores ethics is not a good report, however interesting its findings may be. This chapter states the rules that every school researcher should follow.

Permission and informed consent

Before any data are collected, the researcher needs permission from the head teacher and, where the school requires it, from the education authority. Parents or guardians must be told, in plain language, what the study is about, what their child will be asked to do, how long it will take, and that taking part is voluntary. For most classroom studies, a short letter home, with a form to sign and return, is enough. Pupils themselves should also be told, in words they can understand, that they are taking part in a study, that they may choose not to answer a question, and that their schoolwork will not be affected by their decision. A child who does not want to be interviewed must not be interviewed, and a parent who does not give consent must have that decision respected without any penalty to the child.

Confidentiality and anonymity

Pupils' names must never appear in a report, a presentation or a shared file. Each pupil should be given a code number, and the list linking codes to names should be kept separately and securely, and destroyed when the study is finished. Test scores, interview notes and samples of work should be stored where other pupils cannot see them. Photographs of children should not be taken or used without specific written permission from parents, and even with permission they should be used only for the purpose agreed. When a report quotes a pupil, it should use a code ('Pupil 7') rather than a name, and it should avoid details that would allow a reader in the school to identify the child.

Avoiding harm and being honest about risk

Classroom research should not harm pupils, waste their learning time, or single them out in ways that embarrass them. An intervention that takes a great deal of lesson time must be justified by its likely benefit. A study must not label children in a way that follows them: a report should never describe a child as 'slow', 'weak' or 'a problem', even in private notes, because such labels shape how adults treat children. If a study reveals that a child needs help — for example, a pupil whose reading is far

behind the class — the researcher has a duty to tell the class teacher and to make sure help is offered, rather than simply recording the difficulty as data. Finally, ethical research is honest research: data must not be invented, altered or selectively reported, and the contribution of colleagues and pupils should be acknowledged in the report.

Ethics checklist — complete before collecting data

Head teacher permission obtained. Parents informed in plain language and consent forms returned. Pupils told in simple words and free to decline. Names replaced by codes; code list stored separately. No photographs without specific written permission. Data stored securely. Any child found to need help referred to the class teacher. If any item is not done, do not begin data collection.

Chapter 11

A Complete Worked Example: Improving Reading Fluency in a Grade 5 Class

This chapter is a complete model report. It is clearly labelled as fictional: the school, the pupils and the numbers are invented for teaching purposes, to show what a finished action-research report looks like. A real report would use the writer's own school, pupils and results. The structure, length and style of this example can be imitated; its content must not be copied as if it were real data.

MODEL REPORT — FICTIONAL STUDY

Improving Reading Fluency Through Paired Reading: An Action Research Study in Grade 5

By A. Teacher, Green Valley Primary School | Term 2, 2026 | Fictional example for teaching purposes

Abstract (model)

This study asked whether ten minutes of daily paired reading would improve the reading fluency of Grade 5 pupils over six weeks. Thirty pupils took a one-minute reading fluency test before and after the intervention, and twelve pupils were interviewed. Average fluency rose from 62 to 78 words correct per minute. Twenty-four pupils improved by ten or more words per minute; five showed little gain. Pupils reported feeling less afraid of making mistakes when reading with a partner. The study recommends continuing paired reading with extra support for the five pupils who did not gain, and repeating the study with a second class.

1. Introduction (model)

In Term 1, the Grade 5 class at Green Valley Primary School read aloud slowly and hesitantly. In a class reading assessment, the average fluency was 62 words correct per minute, and eight pupils read fewer than 50 words per minute. Pupils often stopped at unfamiliar words, and several said they disliked reading aloud. The teacher decided to investigate whether paired reading — pupils reading aloud in pairs, helping each other with difficult words — would improve fluency and confidence. The research question was: Does ten minutes of daily paired reading improve the reading fluency of Grade 5 pupils over six weeks, measured by words read correctly per minute?

2. Method (model)

Thirty pupils (16 girls, 14 boys, aged 10 to 11) took part. Paired reading ran for ten minutes at the start of every English lesson, four days a week, for six weeks. Pupils were paired so that a stronger reader worked with a weaker reader, and pairs changed every two weeks. Each pair had a record sheet on which they noted the

book and the pages read. Fluency was measured with a one-minute reading test using a passage of similar difficulty before and after the six weeks. Twelve pupils, chosen to include strong, average and weak readers, were interviewed for five minutes each about how they felt about reading. The head teacher gave permission, parents were informed by letter, and pupils were given code numbers; no names are used in this report.

3. Findings (model)

Average fluency rose from 62 words correct per minute (range 34–91) to 78 (range 41–105), an average gain of 16 words per minute. Table 1 summarises the results by starting level. The largest gains were among pupils who began below 50 words per minute. Five pupils gained between zero and four words; one pupil scored two words lower in the post-test. In interviews, nine of the twelve pupils said they felt less afraid of making mistakes, and eight said they now read at home more often. Three pupils said they found the activity boring by the fifth week, and two said their partner read too fast for them.

Starting level	Pupils	Pre-test avg.	Post-test avg.	Gain
Below 50 wpm	8	43	64	+21
50-70 wpm	13	60	76	+16
Above 70 wpm	9	82	94	+12
Whole class	30	62	78	+16

4. Discussion and recommendations (model)

Paired reading was associated with a clear improvement in fluency, especially for the weakest readers, and with greater confidence. The study cannot prove that paired reading alone caused the improvement, because pupils also had six more weeks of ordinary English lessons and extra reading practice. The five pupils who showed little gain may need a different kind of support, such as individual reading with the teacher. The study recommends: continue paired reading for ten minutes a day; change partners every two weeks and match reading speed more carefully; give the five pupils who did not gain extra individual support; and repeat the study with the parallel Grade 5 class next term, to see whether the result is repeated.

This model report is about 700 words, excluding tables. A real report would be longer, with a literature review, fuller discussion and appendices holding the test, the interview questions and the record sheet. Its value as a model is its structure: question, method, evidence, honest discussion, and recommendations that follow from the findings.

Chapter 12

Common Mistakes and a Final Checklist

Most weak research reports fail in the same few ways. Knowing those mistakes in advance makes them easy to avoid. This chapter lists the most common errors, explains why each one matters, and ends with a final checklist to use before a report is submitted or shared.

Common mistakes

1. Starting without a clear question. The study collects a great deal of data and then the writer cannot say what it was for. Fix it by writing the question down before collecting anything, and checking every instrument against it.

2. Collecting too much data. A questionnaire of forty questions, interviews with thirty pupils, and three different tests produce a pile that cannot be analysed in the time available. Fix it by choosing two or three methods and keeping each one short.

3. Confusing opinion with evidence. 'The method worked brilliantly' is not a finding. Fix it by following every claim with a number, a table or a quotation.

4. Reporting only the good news. Pupils who did not improve, activities that pupils disliked, and results that contradict the main finding are left out. Fix it by reporting the full range of results, including the disappointing ones.

5. Claiming too much. A study of one class over six weeks is described as proving that a method works for all children. Fix it by matching the claim to the evidence and stating the limits openly.

6. Vague methodology. The report does not say how many pupils took part, how long the study lasted, or how data were analysed, so no reader can judge or repeat the work. Fix it by writing the methodology as if another teacher had to repeat the study from the report alone.

7. Invented or careless references. Sources are listed that were never read, or cited with wrong details. Fix it by reading every source cited and checking every detail against the original.

8. Poor presentation. No page numbers, tables without titles, quotations without sources, and spelling errors throughout. Fix it by proofreading, by asking a colleague to read the report, and by using the checklist below.

Final checklist — before you submit

- The title names the topic, the method and the setting.
- The abstract (150–250 words) states the question, method, main findings and recommendation.
- The research question is clear, focused and answerable with the evidence

collected.

- The methodology is detailed enough for another teacher to repeat the study.
- Every claim in the findings is supported by a number, a table, a chart or a quotation.
- Results for pupils who did not improve are reported, not hidden.
- The discussion considers alternative explanations and states the limits of the study.
- Every recommendation follows from a finding in the report.
- Ethical steps — permission, consent, confidentiality — are described.
- Every source in the reference list is cited in the text, and every citation is in the list.
- Tables and charts are numbered, titled and mentioned in the text.
- Names of pupils are replaced by codes; no child can be identified.
- The report has been proofread, and a colleague has read it.
- Appendices hold the questionnaire, test and observation sheets used.

Chapter 13

Conclusion

Education research is not a distant academic activity. It is a practical craft that any teacher can learn: ask a clear question about your own classroom, gather evidence carefully, analyse it honestly, and write it up so that others can learn from it. The studies that matter most in a school are rarely the largest or the most complicated. They are the small, honest studies that answer a real question and change what happens on Monday morning.

This report has described the main types of education research, explained the Plan–Act–Observe–Reflect cycle of action research, shown how to formulate a research question, collect data with tests, questionnaires, interviews and observation, and analyse that data simply and honestly. It has given the standard structure of a research report, guidance on writing findings as evidence rather than opinion, advice on using findings to improve teaching, and the ethical rules that must govern all research with children. The worked example in Chapter 11 shows what a finished report looks like, and the checklist in Chapter 12 gives the writer a final test before submitting.

The most important advice in this report can be stated in one sentence: be honest. Report what you did, report what you found — including what disappointed you — and claim only what your evidence supports. A teacher who does that, study after study, becomes a better teacher and helps her colleagues become better teachers too. That is the real purpose of education research, and it is within the reach of every classroom.

Begin small. Choose one question that matters to you, plan one cycle of action research, and write it up. Then share it with a colleague and begin again. Over a year, a teacher who does this will know more about her pupils, her methods and her own teaching than any amount of reading alone could teach her — and she will have the written evidence to prove it.

Suggested Reading

The following are general, well-known works and resources that teachers beginning education research commonly find useful. Readers should consult the editions available in their own library or institution; no specific page or chapter is claimed here beyond the general scope of each work.

- Books on action research and teacher inquiry — introductory texts on the Plan–Act–Observe–Reflect cycle and on carrying out small-scale classroom studies.
- Books on research methods in education — general introductions to quantitative and qualitative methods, including surveys, interviews, observation and case study design.
- Books on academic writing for education — guides to structuring a report or dissertation, writing findings, and presenting tables and figures clearly.
- Guides to research ethics with children — general guidance on consent, confidentiality and safeguarding in school-based research, including institutional ethics guidelines.
- Official guidance from education ministries and examination bodies in the reader's own country — curriculum documents, assessment guidance and school research policies, which vary by country and should always be checked locally.
- University open-access introductions to educational research — freely available online course materials and library guides on formulating questions, collecting data and analysing results simply.

A note on references: a research report should cite only sources the writer has actually read. General reading lists such as this one are a starting point for learning, not a substitute for the reference list of a finished report, which must give full and accurate details of every work cited in the text.